**LAB EXPERIMENT RECORD**

**22-10-2026---Set 25: Website Traffic Analysis**

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|---|---|
| **Department** | AIML |
| **Course Code / Course Name** | DSA0601 / Data Handling and Visualization |
| **Experiment** | Data Visualization using R and Tableau |
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| **Date** | 22 OCTOBER 2026 |
| **Day** | THURSDAY |

**Aim**

To analyze website traffic data in R and Tableau using a line chart, bar chart, stacked area chart and an interactive dashboard.

**Dataset: Website Traffic**

| Date | Page Views | Click-through Rate |
|---|---|---|
| 2023-01-01 | 1500 | 2.3% |
| 2023-01-02 | 1600 | 2.7% |
| 2023-01-03 | 1400 | 2.0% |
| 2023-01-04 | 1650 | 2.4% |
| 2023-01-05 | 1800 | 2.6% |

**Question 1: Line Chart of Page Views (Tableau)**

**Question: In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.**

**Aim:** To visualize the trend in daily page views using a Tableau line chart with labelled axes and a title.



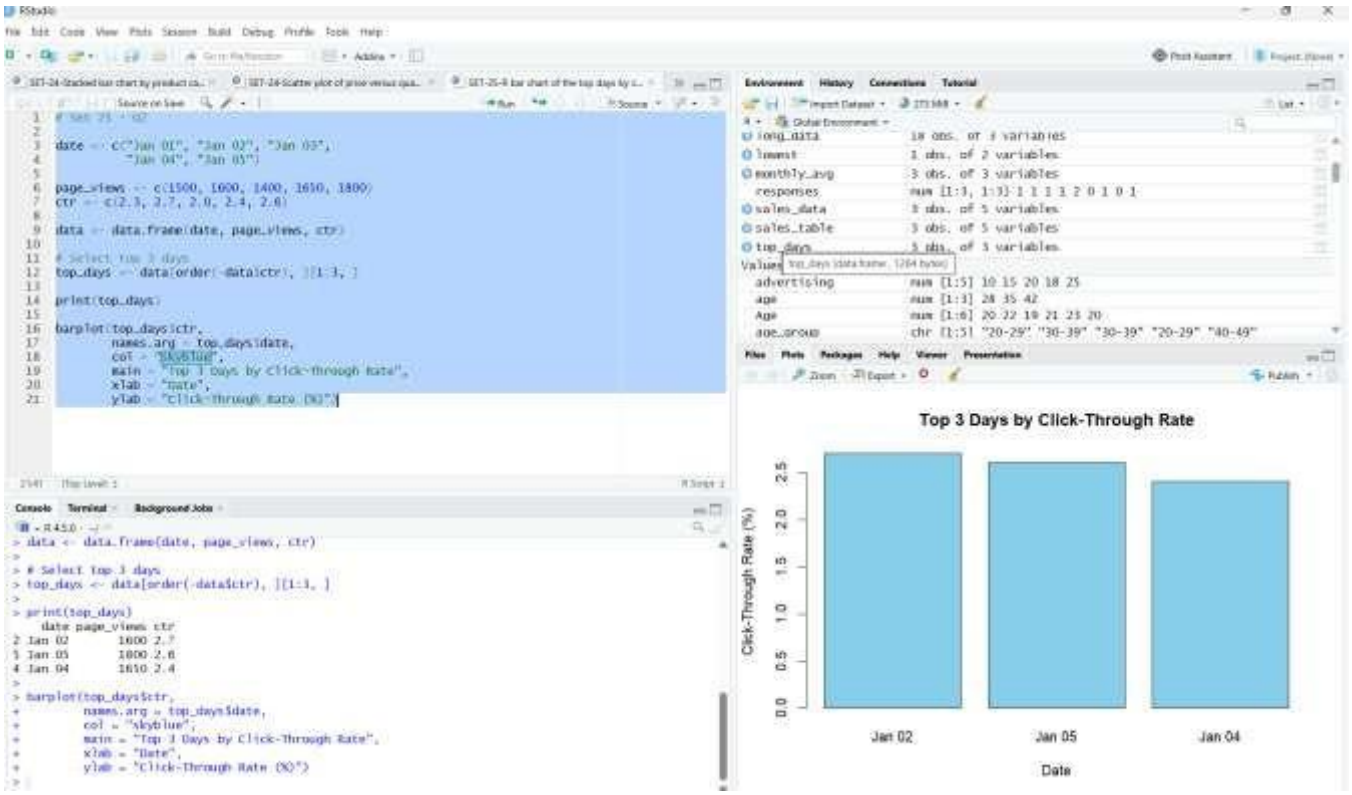

**Result:** Page views rise from 1500 to 1600, dip to 1400 on 2023-01-03, and then climb to 1650 and a peak of 1800 on 2023-01-05, so the overall trend is upward.

**Question 2: Bar Chart of Top N Days by Click-through Rate (R)**

**Question: In R, generate a bar chart to show the top N days with the highest click-through rates.**

**Label the chart elements.**

**Aim:** To identify the top N days with the highest click-through rates using a labelled bar chart.

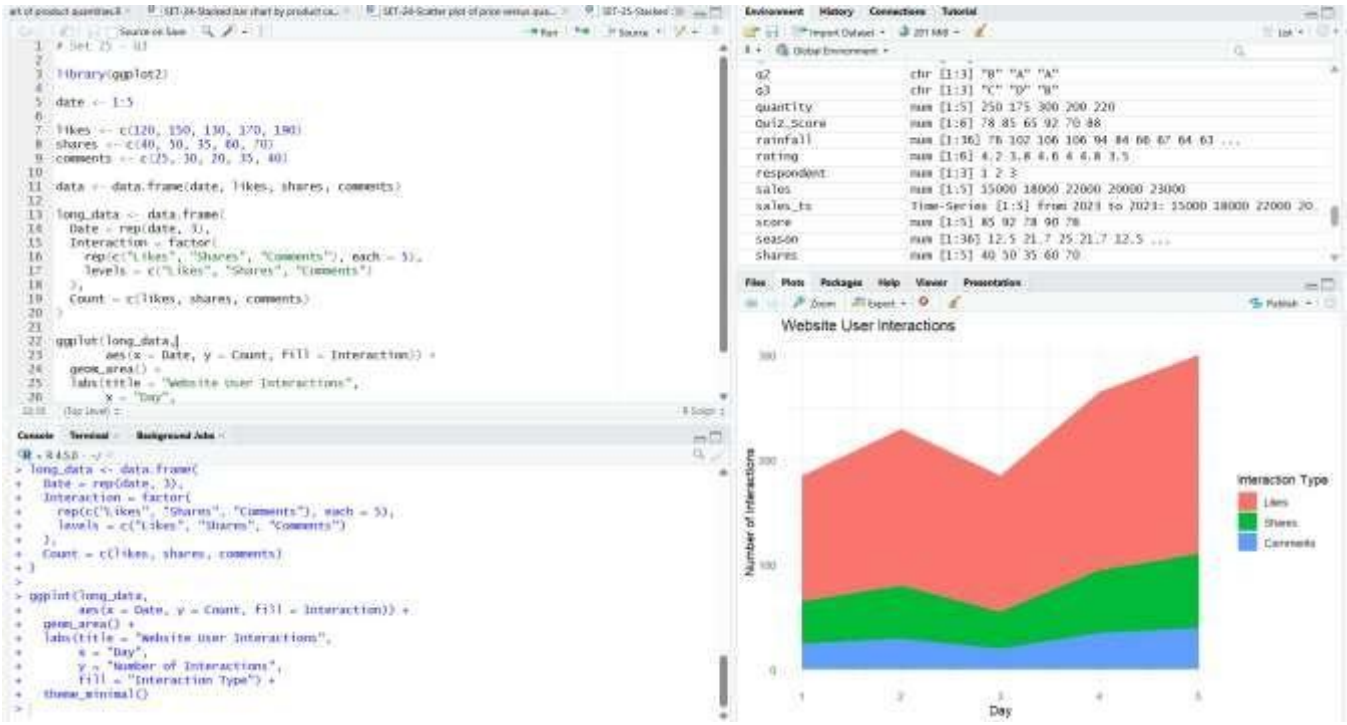



**Result:** For N = 3, 2023-01-02 ranks first (2.7%), followed by 2023-01-05 (2.6%) and 2023-01-04 (2.4%).

**Question 3: Stacked Area Chart of User Interactions (R)**

**Question: In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.**

**Aim:** To display how likes, shares and comments add up to total user interactions over time using a stacked area chart.




**Result:** The chart shows each interaction type as a layer stacked over the days; the dataset has no likes, shares or comments columns, so these values must be supplied.

**Question 4: Tableau Dashboard with Map and Bar Chart**

**Question: In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.**

**Aim:** To build an interactive Tableau dashboard that shows traffic sources on a map and page views by source in a bar chart.




**Result:** The map and bar chart are linked, so selecting a source on the map highlights its page views in the bar chart; the dataset has no source or location column, so these must be supplied.

**Overall Result**

**Observation:** The average page views are 1590 per day and the average click-through rate is 2.4%, with the most views on 2023-01-05 (1800) and the best click-through rate on 2023-01-02 (2.7%).